

HTML: Block & Inline Elements, Attributes, and Layouts

HTML elements can be broadly categorized into **Block-level** and **Inline-level** elements. These, along with **attributes**, **headings**, **paragraphs**, **tables**, **lists**, and **layout tags**, are fundamental for structuring web pages.

Block and Inline Elements

Block-level Elements

- **Characteristics:**
 - Take up the **entire width** of the parent container.
 - Always start on a **new line**.
 - Can contain **other block-level and inline elements**.
- **Common Examples:**

```
<div>, <p>, <h1> ... <h6>, <section>, <article>, <header>, <footer>, <aside>, <main>.
```

- **Use Cases:**
 - Structuring **sections** of a page.
 - Wrapping multiple elements together.
 - Creating **containers** for styling via CSS.

Code Example:

```
<div>
  <h1>Main Heading</h1>
  <p>This paragraph is block-level and starts on a new line.</p>
</div>
```

Inline Elements

- **Characteristics:**

- Take only the **width required** by the content.
- Do **not** start on a new line.
- Usually used for text formatting or small pieces of content.

- **Common Examples:**

```
<span>, <a>, <img>, <strong>, <em>, <mark>, <code>.
```

- **Use Cases:**

- Highlighting text.
- Embedding links or images within text.
- Styling specific words or phrases.

Code Example:

```
<p>This is <span style="color:red;">inline text</span> inside a  
paragraph.</p>
```

Comparison Table

Feature	Block Element	Inline Element
Starts on new line?	Yes	No
Width	Full width of container	Only as wide as content
Can contain	Block & inline	Only inline
Examples	<div>, <p>, <section>	, <a>,

Attributes

Attributes **provide additional information** about HTML elements and modify behavior or appearance.

Common Attributes and Use Cases:

Attribute	Description	Example
id	Unique identifier for an element	<code><p id="para1">Hello</p></code>
class	CSS class for styling multiple elements	<code><div class="container"> Content </div></code>
style	Inline CSS styles	<code><p style="color:blue;">Blue text</p></code>
src	Source for images, scripts	<code></code>
href	URL for links	<code>Google</code>
alt	Alternative text for images	<code></code>
title	Tooltip text	<code><p title="Hover me">Text</p></code>
value	Input value in forms	<code><input value="Hello"></code>
selected	Pre-selected dropdown option	<code><option selected>Apple</option></code>

disabled	Disables a form element	<input disabled>
placeholder	Placeholder text for inputs	<input placeholder="Enter name">

Example:

```
<input type="text" id="username" class="input-field" placeholder="Enter Name">
```

Headings

- HTML supports **6 levels of headings** <h1> to <h6>.
- Use **hierarchically** for SEO and accessibility.
- **Block-level by default.**

Example:

```
<h1>Main Heading</h1>  
<h2>Subheading</h2>  
<h3>Section Heading</h3>  
<h4>Subsection</h4>  
<h5>Smaller Heading</h5>  
<h6>Minor Heading</h6>
```

Best Practices:

- Only **one <h1> per page** (main topic).
 - Use headings to create **logical page structure.**
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Paragraphs

- `<p>` defines a **paragraph of text**.
- Can contain **inline elements** for formatting.

Example:

```
<p>This is a <strong>paragraph</strong> with <em>emphasized</em> and  
<mark>highlighted</mark> text.</p>
```

Attributes:

- `align` → Align text (`left`, `center`, `right`) (*deprecated, use CSS instead*)

Tables

HTML tables are used to **display structured data in rows and columns**. They are widely used for reports, schedules, comparison charts, etc.

Basic Structure

Tag	Purpose
<code><table></code>	Container for the table
<code><caption></code>	Table title or description
<code><tr></code>	Table row

<code><th></code>	Table header (bold & centered by default)
<code><td></code>	Table cell (data)
<code><thead></code>	Groups header rows
<code><tbody></code>	Groups body rows
<code><tfoot></code>	Groups footer rows

Example:

```

<table border="1">
  <caption>Student Marks</caption>
  <thead>
    <tr>
      <th>Name</th>
      <th>Math</th>
      <th>Science</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>John</td>
      <td>95</td>
      <td>88</td>
    </tr>
    <tr>
      <td>Jane</td>
      <td>89</td>
      <td>92</td>
    </tr>
  </tbody>

```

```
<tfoot>
  <tr>
    <td>Total</td>
    <td>184</td>
    <td>180</td>
  </tr>
</tfoot>
</table>
```

Important Table Attributes

Attribute	Purpose	Example
<code>border</code>	Border thickness	<code><table border="2"></code>
<code>cellpadding</code>	Space inside cells	<code><table cellpadding="10"></code>
<code>cellspacing</code>	Space between cells	<code><table cellspacing="5"></code>
<code>colspan</code>	Merge columns	<code><td colspan="2">Merged</td></code>
<code>rowspan</code>	Merge rows	<code><td rowspan="2">Merged</td></code>
<code>align</code>	Alignment of cell (deprecated)	<code><td align="center">Text</td></code>

Nested Tables

Tables can be nested for **complex layouts**:

```
<table border="1">
  <tr>
    <td>
      <table border="1">
        <tr><td>Nested 1</td></tr>
        <tr><td>Nested 2</td></tr>
      </table>
    </td>
    <td>Main Table Cell</td>
  </tr>
</table>
```

Best Practices for Tables

- Use `<thead>`, `<tbody>`, `<tfoot>` for **readability and accessibility**.
 - Avoid using tables for **layout purposes**; use CSS instead.
 - Always provide **captions** to describe the table.
 - Use **semantic** `<th>` for headers instead of styling `<td>` bold.
-

Dropdowns (Select Menu)

- Created with `<select>` and `<option>` elements.
- Allows **users to select one or multiple options**.

Example:

```
<label for="fruits">Choose a fruit:</label>
<select id="fruits" name="fruits">
  <option value="apple">Apple</option>
  <option value="banana">Banana</option>
  <option value="mango" selected>Mango</option>
</select>
```


Attributes:

- `multiple` → Allow multiple selections
- `disabled` → Disable dropdown

Quotations

Inline Quotes → `<q>`

- Automatically adds **quotation marks**.

```
<p>He said, <q>HTML is fun!</q></p>
```

Block Quotes → `<blockquote>`

- Used for **long quotations**.
- Optional `cite` attribute for source URL.

```
<blockquote cite="https://www.example.com">  
  HTML is the backbone of every website.  
</blockquote>
```

Lists

HTML lists are used to **group related items** in an organized way. Lists can be **ordered**, **unordered**, or **description lists**.

Unordered List `<ul>`

- Used for items with **no specific order**.
- Items are wrapped in `<li>` tags.
- Can be **styled with bullets, squares, circles** via CSS.

Example:

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Attributes:

- **type** → Bullet style (disc, circle, square) (*deprecated, use CSS instead*)

Ordered List ****

- Used for **numbered or sequential items**.
- Items wrapped in ****.

Example:

```
<ol>
  <li>Install VS Code</li>
  <li>Create project folder</li>
  <li>Run live server</li>
</ol>
```

Attributes:

- **type** → Numbering style (1, a, A, i, I)
- **start** → Starting number
- **reversed** → Reversed order

Example with Attributes:

```
<ol type="A" start="3" reversed>
  <li>Step C</li>
  <li>Step B</li>
  <li>Step A</li>
</ol>
```

Description List <dl>

- Used for **term-description pairs**.
- Tags: <dl> (list), <dt> (term), <dd> (description).

Example:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Nested Lists

- Lists can be **nested inside another list** to create hierarchy.

Example:

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Mango</li>
    </ul>
  </li>
  <li>Vegetables
    <ul>
      <li>Carrot</li>
      <li>Spinach</li>
    </ul>
  </li>
</ul>
```

Lists in Forms and Menus

- Dropdowns often **use for custom menus** in navigation.
- Nested lists are useful for **multi-level menus**.

Example:

```
<nav>
  <ul>
    <li>Home</li>
    <li>About</li>
    <li>Services
      <ul>
        <li>Web Development</li>
        <li>App Development</li>
      </ul>
    </li>
    <li>Contact</li>
  </ul>
</nav>
```

Best Practices for Lists

- Use **** for **unordered**, **** for **ordered**, **<dl>** for **definitions**.
 - Always wrap **list items** with ****.
 - Avoid excessive nesting for readability.
 - Use **CSS** to style bullets, numbers, and indentation.
 - Nested lists are ideal for **navigation menus and subtopics**.
-

Blocks

Generic container elements used to structure content.

Common Elements:

Element	Purpose
<code><div></code>	Generic block container
<code><section></code>	Group related content
<code><article></code>	Independent content (blog, news)
<code><header></code>	Top section of page
<code><footer></code>	Bottom section
<code><aside></code>	Sidebar content

Example:

```
<section>
  <h2>About Me</h2>
  <p>I am a web developer specializing in frontend development.</p>
</section>
```

Layout Elements

- **Semantic elements** improve **readability, accessibility, and SEO**.

Common Layout Elements:

- `<header>` → Page header with logo/navigation
- `<nav>` → Navigation menu
- `<main>` → Main page content
- `<section>` → Section of content
- `<article>` → Independent content block
- `<aside>` → Sidebar or supplementary content
- `<footer>` → Page footer with copyright/contact

Example:

```
<body>
  <header>
    <h1>Website Header</h1>
    <nav>
      <a href="#">Home</a> | <a href="#">About</a>
    </nav>
  </header>

  <main>
    <article>
      <h2>Blog Post</h2>
      <p>Content of the blog post goes here.</p>
    </article>
    <aside>
      <h3>Related Links</h3>
      <p>See more tutorials.</p>
    </aside>
  </main>

  <footer>
    <p>(c) 2025 My Website</p>
  </footer>
</body>
```